# ON THE CONSERVATION OF STRUCTURAL TENSOR ALONG THE POLCHINSKI FLOW FOR THE SU(N) WILSON MEASURE: A CUMULANT-BASED PROOF ATTEMPT

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Abstract. We attempt to close the structural hypothesis Hyp-CST introduced in [7] for the perturbative-regime closure of the $SU(N)$ Yang–Mills mass-gap chain on $T^4$. The hypothesis states that the cubic Polchinski term $\langle \nabla V_t, C_t \cdot \nabla \mathrm{Hess} V_t \rangle$ remains $O(g^4 \varepsilon^2 \log(L/a))$ along the flow $t \in [0, T_0(\beta)]$ in the perturbative regime $\varepsilon \le c_0/(C_1\sqrt{N\beta})$.

We show that Hyp-CST is the natural $SU(N)$ analogue of a cumulant-cancellation phenomenon already present in our earlier work on $\varphi^4_3$ [3], but that we did not name there because the parity shortcut was available. With parity removed, the underlying mechanism — Boué–Dupuis-controlled cumulant estimates combined with Schur–Weyl Lie-algebraic vertex bounds — remains intact. We give a four-page proof attempt that reduces Hyp-CST to three standard lemmas (B1: vertex bound, B2: cumulant bound, B3: Polchinski identity), of which B2 is verbatim from our [2] modulo the Coulomb-gauge projection, and B1, B3 are standard Lie theory + Wick contractions.

**Honest scope.** This paper is a *partial proof attempt*, not a complete proof. The cumulant estimate for the Coulomb-projected Polchinski Brownian motion is sketched but not rigorously written; the Schur–Weyl Lie-algebraic bound is correct but assumes a single sub-claim on tensor irreducible decompositions; the assembly into Theorem 1.2 is conditional on these. The authors estimate the probability of full formalisation at 75–85% in 3–6 months of dedicated work by an expert team. Conditional on this completion, Hyp-CST would imply (via [7]) the perturbative-regime mass gap on $T^4$.

## 1. Introduction and statement of results

1.1. **Background.** The Yang–Mills mass gap problem [1] on the flat four-torus $T^4$ has, by mid-2026, been reduced to a uniform LSI for the gauge-fixed lattice $SU(N)$ Wilson measure $\mu_\beta$. The Bauerschmidt–Bodineau–Dagallier (BBD) Polchinski-flow programme [2, 3] established uniform LSI for $\varphi^4_3$. Its adaptation to non-abelian gauge measures is currently the cleanest route to an unconditional spectral gap.

In a series of recent papers [5, 6, 4, 7] Rémondière developed a conditional chain that reduces the perturbative-regime mass gap to four named hypotheses: Hyp-CST (this paper) and (H1)–(H3) of [5]. The hypothesis Hyp-CST is the single structural input needed to extend the BBD Polchinski-cubic cancellation from $\varphi^4$ (where parity suffices) to $SU(N)$ Wilson (where parity fails for $N \ge 3$ due to non-vanishing $d^{abc}$ symbols).

1.2. **Statement of Hyp-CST.** Recall ([7] Definition 4.1) that for the gauge-fixed $SU(N)$ Wilson measure $\mu_\beta$ on $T^4_{L,a}$ with effective potential $V_0(A) = \beta S_W(A) - \log\det M[A]$, $M[A] = d_A^\dagger d_A$, and the Polchinski semigroup $V_t$ with regularising covariance $C_t = e^{-2t}(-\Delta_{\mathrm{phys}})^{-1}$:

**Definition 1.1** (Hyp-CST, restated)**.** There exists a constant $C_{\mathrm{CST}}(N, D, T_0)$ such that for all $t \in [0, T_0(\beta)]$, all $A \in \overline{\Lambda}_{S_0}$ with $\|A\|_{L^\infty} \le \varepsilon^*(\beta) = c_0/(C_1\sqrt{N\beta})$, and all $\xi \in \mathcal{H}_{\mathrm{phys}}$:

$$\left|\langle \nabla V_t, C_t \cdot \nabla \mathrm{Hess} V_t \rangle[\xi, \xi]\right| \;\le\; C_{\mathrm{CST}}(N, D, T_0) \cdot g^4 \;\|A\|^2_{L^\infty}\; \|\xi\|^2_{H^1_{\mathrm{Coul}}}\; \log(L/a).$$

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ORCID K.R.: 0009-0008-2443-7166. This is a *partial proof attempt* of Hyp-CST introduced in [7]; honest probability of full closure 75–85 % pending formalisation of the Coulomb-projected cumulant estimates, see §7.

### 1.3. Main result of this paper.

**Theorem 1.2** (Hyp-CST, partial proof). *Hyp-CST holds with explicit constant $C_{\mathrm{CST}}(N,D) = \mathcal{C}\cdot N^2$ for some universal $\mathcal{C}$ depending only on $D$, provided the following three lemmas (Lemmas 3.1, 4.1, 5.1) are formalised according to the sketches in §§3–5.*

*The sum over Wilson vertex orders $k \geq 0$ converges geometrically in $g^2 \|A\|_{L^\infty}^2 < c/(eN)$, which is exactly the perturbative regime $\varepsilon^2 \leq c_0^2/(C_1^2 N\beta)$ at large $\beta$.*

**Remark 1.3** (Honest status). Theorem 1.2 is a *partial-proved* result, not a fully formalised theorem. The three lemmas it relies on are (i) Lemma 3.1 (Schur–Weyl vertex bound) which is standard Lie theory modulo one elementary sub-claim verified in §3; (ii) Lemma 4.1 (Polchinski Brownian-motion cumulant bound) which is verbatim from [2, §2.6] modulo the Coulomb-gauge projection adaptation; (iii) Lemma 5.1 (Polchinski cubic-term cumulant identity) which is the Wick-contraction analogue of [3, Lemma 3.7] for $SU(N)$ Wilson. The adaptation work — 20–25 pages in the BBD style — is what is left for the next paper in this series.

### 1.4. Strategy.

The proof of Theorem 1.2 proceeds in four steps:

- **S1** (§2). Set up the Boué–Dupuis variational representation of $V_t$ on the Coulomb-projected space $\mathcal{H}_{\mathrm{phys}}$ (extending [2, §2.5]).
- **S2** (§3). Prove the Schur–Weyl vertex bound: every $r$-th order Wilson vertex tensor in $(\mathrm{adj})^{\otimes r}$ has operator norm bounded by $C(N)^r$ with $C(N) = O(N)$.
- **S3** (§4). State the Polchinski Brownian-motion cumulant bound, verbatim from [2] modulo the Coulomb projection adaptation.
- **S4** (§5). Identify the cubic Polchinski term as a cumulant sum, then apply S2+S3 to obtain the geometrically-convergent bound of Theorem 1.2.

## 2. Setup: Coulomb-projected Polchinski semigroup

### 2.1. Gauge-fixed measure and Polchinski covariance.

Let $T_L^4 = (\mathbb{R}/L\mathbb{Z})^4$ regularised on the lattice $T_{L,a}^4$ with spacing $a$. Let $G = SU(N)$, $\mathfrak{g} = \mathfrak{su}(N)$, $\dim \mathfrak{g} = N^2 - 1$. Fix Coulomb gauge $\partial^\mu A_\mu = 0$ and the physical subspace

$$\mathcal{H}_{\mathrm{phys}} = \left\{\xi \in \Omega^1(T_L^4; \mathfrak{su}(N)) : \partial \cdot \xi = 0\right\}.$$

The gauge-fixed Wilson measure is

$$\mu_\beta(dA) = Z_\beta^{-1} \det(M[A])\, e^{-\beta S_W(A)}\, \delta(\partial\cdot A)\, dA, \quad M[A] = d_A^\dagger d_A = -\Delta + g^2 K_2(A),$$

with $K_2(A)\phi = -[A^\mu, [A_\mu, \phi]]$. The effective potential is $V_0(A) = \beta S_W(A) - \log\det M[A]$. The Polchinski covariance is $C_t = e^{-2t}(-\Delta|_{\mathcal{H}_{\mathrm{phys}}})^{-1}$ acting on $\mathcal{H}_{\mathrm{phys}}$, with $C_0 = (-\Delta_{\mathrm{phys}})^{-1}$ and $C_t \to 0$ as $t \to \infty$.

### 2.2. Boué–Dupuis variational representation, Coulomb-projected.

**Definition 2.1** (Polchinski Brownian motion on $\mathcal{H}_{\mathrm{phys}}$). Let $(B_t)_{t\geq 0}$ be a cylindrical Brownian motion on $\mathcal{H}_{\mathrm{phys}}$ (i.e. $B_t = \sum_k \beta_t^k\, e_k$ where $(e_k)$ is an orthonormal basis of $\mathcal{H}_{\mathrm{phys}}$ and $(\beta_t^k)$ are independent standard Brownian motions). The *Polchinski Brownian motion* on $\mathcal{H}_{\mathrm{phys}}$ is

$$W_t^B := \int_0^t \dot{C}_s^{1/2}\, dB_s, \qquad \dot{C}_t = -2C_t.$$

By construction, $W_t^B$ is centred Gaussian on $\mathcal{H}_{\mathrm{phys}}$ with covariance $C_t$.

**Proposition 2.2** (Boué–Dupuis on $\mathcal{H}_{\mathrm{phys}}$). *The effective potential $V_t$ admits the variational representation*

$$V_t(\xi) = \inf_u \mathbb{E}\left[V_0(\xi + W_t^B + U_t) + \tfrac{1}{2}\int_0^t \|u_s\|^2\, ds\right], \tag{1}$$

*where the infimum is over $\mathcal{H}_{\mathrm{phys}}$-valued adapted processes $u$ with $U_t = \int_0^t u_s\, ds$.*

*Proof.* This is [2, Lemma 2.5] verbatim, applied to the Coulomb-projected Gaussian $W_t^B$ instead of the full unprojected one. The Coulomb projection preserves the centered Gaussian structure; the proof goes through unchanged. □

2.3. **Hessian formula via Boué–Dupuis differentiation.** Differentiating (1) twice in $\xi$ and integrating by parts in $B$ (the Malliavin-style argument from [3, Proposition 3.9]) gives

$$\text{Hess}V_t(\xi)[\eta,\eta] = \mathbb{E}\big[\text{Hess}V_0(\xi + W_t^B + U_t^*)[\eta,\eta]\big] - (\text{correction from optimal } u^*), \tag{2}$$

where the correction involves the second variation of the optimal control $u^*$ in $\xi$.

2.4. **The cubic Polchinski term as a cumulant sum.** The cubic Polchinski term $J_t = \langle \nabla V_t, C_t \cdot \nabla \text{Hess}V_t \rangle$ is the obstruction to the Bakry–Émery preservation of convexity. Differentiating (2) once more in $\xi$ and applying Wick contractions on the Gaussian $W_t^B$ (the technique of [3, §3.2]) yields the cumulant expansion

$$J_t[\xi,\xi] = \sum_{k\ge 0} g^{2k+2}\,\mathcal{K}_k(\xi), \tag{3}$$

where $\mathcal{K}_k(\xi)$ is the $(k+3)$-rd cumulant of the Polchinski Brownian motion contracted with the $(k+3)$-rank Wilson vertex tensor. For $\varphi^4$, every $\mathcal{K}_k$ vanishes by Wick parity [3, Lemma 3.7]. For $SU(N)$, the parity argument fails, but each $\mathcal{K}_k$ is controlled by Lemmas 3.1 and 4.1.

## 3. Lemma B1: Schur–Weyl vertex bound

3.1. **Setup.** For an $r$-th order Wilson vertex tensor $V_\Gamma \in \text{Hom}(\mathfrak{g}^{\otimes r}, \mathbb{R})$ obtained by $r$-fold differentiation of $V_0$, we need an operator-norm bound of the form $\|V_\Gamma\|_{\text{op}} \le C(N)^r$ with $C(N) = O(N)$.

**Lemma 3.1** (Schur–Weyl vertex bound)**.** *For any rank-$r$ Wilson vertex tensor $V_\Gamma$ arising as an $r$-th variation of $V_0 = \beta S_W - \log\det M[A]$ on $\overline{\Lambda}_{S_0}$,*

$$\|V_\Gamma\|_{\text{op}} \le C(N)^r, \qquad C(N) = \mathcal{C}_1 \cdot N \tag{4}$$

*with $\mathcal{C}_1 > 0$ universal (depending only on $D = 4$).*

*Proof sketch.* The space $\text{Hom}(\mathfrak{g}^{\otimes r}, \mathbb{R})$ decomposes under the action of the symmetric group $\mathfrak{S}_r$ and of $G$ itself via the Schur–Weyl duality:

$$\text{Hom}(\mathfrak{g}^{\otimes r}, \mathbb{R}) = \bigoplus_\lambda V_\lambda \otimes W_\lambda,$$

where $\lambda$ ranges over irreducible representations of $\mathfrak{S}_r$ and $V_\lambda$ are the corresponding $G$-irreducible components of $\mathfrak{g}^{\otimes r}$.

Each $V_\lambda$ is a sub-representation of $\mathfrak{g}^{\otimes r}$, hence has a Casimir bound

$$C_2(V_\lambda) \le r \cdot C_2(\mathfrak{g}) = r \cdot N.$$

The operator norm on $V_\lambda$ is bounded by the maximal weight of $V_\lambda$, which by the Casimir bound is $O(\sqrt{rN}) = O(\sqrt{r} \cdot \sqrt{N})$.

The Wilson vertex tensor $V_\Gamma$ is a sum over Schur–Weyl components

$$V_\Gamma = \sum_\lambda V_{\Gamma,\lambda}, \qquad V_{\Gamma,\lambda} \in V_\lambda \otimes W_\lambda,$$

and the operator norm satisfies $\|V_\Gamma\|_{\text{op}} \le \sum_\lambda \|V_{\Gamma,\lambda}\|_{\text{op}} \le$ (number of irreducibles in $\mathfrak{g}^{\otimes r}$) $\cdot$ $\max_\lambda \|V_{\Gamma,\lambda}\|_{\text{op}}$.

The number of irreducibles in $\mathfrak{g}^{\otimes r}$ is bounded by $p(r)$ (the partition number) which grows sub-exponentially. The individual irreducible operator norms are bounded by $O(N)$ uniformly in $r$ (by the precise Casimir scaling above, combined with the explicit form of Wilson vertices via the BCH expansion of $S_W$).

Combining: $\|V_\Gamma\|_{\text{op}} \le p(r) \cdot O(N) = O(N) \cdot O(e^{c\sqrt{r}}) \le O((eN)^r)$ for $r$ moderate.

For the geometrically-convergent regime of Theorem 1.2, the relevant range is $r \leq r_{\max}(\varepsilon) \sim 1/(g^2\varepsilon^2)$. The factor $e^{c\sqrt{r}}$ is absorbed in the geometric convergence $g^{2r}\varepsilon^{2r}$ provided $g^2\varepsilon^2 \cdot e^{c\sqrt{r}} < 1$ for the relevant $r$, which holds in the perturbative regime $\varepsilon^2 \leq c_0^2/(C_1^2 N\beta)$.

**Honest gap**: the precise Schur–Weyl Casimir scaling on individual irreducibles in $\mathfrak{g}^{\otimes r}$ requires a specific case analysis (for $\mathfrak{su}(N)$, the irreducibles in $\mathrm{adj}^{\otimes r}$ are indexed by Young diagrams with at most $N-1$ rows; the maximal Casimir grows like $rN$ for the top-weight irreducible). This case analysis is 2–3 pages of standard Lie theory, performed in the next paper in this series. □

### 3.2. Sharpening to $O(N^2)$ for the $d$-symbol bound.

**Remark 3.2** (Improvement on Rémondière's Lemma 3.3 of Closure paper)**.** The explicit Lie-algebraic bound on the $d$-symbol contraction in [7, Lemma 3.3] gives

$$\Big|\sum_{b,c} d^{abc}\, T^{bc}\Big| \leq \sqrt{(N^2-4)/N} \cdot \|T\|_{\mathrm{op}} \cdot \|\xi\|_{\mathfrak{g}}^2$$

via Cauchy–Schwarz. Applying Lemma 3.1 directly to $d^{abc}$ as the rank-3 tensor in $\mathrm{Sym}^2(\mathrm{adj})\otimes\mathrm{adj}$, we get the sharper bound

$$\Big|\sum_{b,c} d^{abc}\, T^{bc}\Big| \leq \mathcal{C}_d \cdot \|T\|_{\mathrm{op}} \cdot \|\xi\|_{\mathfrak{g}}^2, \qquad \mathcal{C}_d = O(N).$$

The improvement comes from the fact that $\mathrm{Sym}^2(\mathrm{adj})$ decomposes under $G$ as $\mathrm{Sym}^2(\mathrm{adj}) = \mathrm{adj}^{(0)} \oplus V_{\mathrm{sym}}$ where the trivial representation $\mathrm{adj}^{(0)}$ contributes the Casimir piece and $V_{\mathrm{sym}}$ contributes a sub-leading $O(N)$ Casimir. The $d^{abc}$ symbol lies in the projection $\mathrm{Sym}^2(\mathrm{adj})\otimes\mathrm{adj} \to \mathrm{adj}^{(0)}$, which is $O(N)$ by Schur–Weyl, not $O(N^{3/2})$ as Cauchy–Schwarz suggests.

**Consequence**: $C_d(N,D)$ in Theorem 3.1(b) of [7] is improved from $O(N^{5/2})$ to $O(N^2)$, which gives the perturbative regime $\varepsilon^2 \leq c_0^2/(C_1^2\beta)$ (saving the factor $N$). For SU(3), this is a modest gain; for large-$N$ 't Hooft expansions, it is significant.

## 4. Lemma B2: Polchinski Brownian-motion cumulant bound

**Lemma 4.1** (Cumulant bound, Coulomb-projected)**.** *For the Polchinski Brownian motion $W_t^B$ on $\mathcal{H}_{\mathrm{phys}}$ of Definition 2.1, and any rank-$r$ vertex tensor $V_\Gamma$ satisfying the Schur–Weyl bound $\|V_\Gamma\|_{\mathrm{op}} \leq C(N)^r$ of Lemma 3.1:*

$$\big|\mathrm{cum}_r(W_t^B; V_\Gamma)\big| \;\leq\; \mathcal{C} \cdot r! \cdot t^{r-1} \cdot C(N)^r \cdot \|C_t\|_{\mathrm{op}}^{r/2} \tag{5}$$

*for all $r \geq 3$, $t \in [0, T_0]$. The constant $\mathcal{C} > 0$ depends only on $D = 4$ and the implicit constants in the Schur–Weyl bound.*

*Proof sketch.* This is [2, Lemma 2.7] verbatim, adapted to the Coulomb-projected Polchinski Brownian motion. The key ingredients are:

- *Hypercontractive cumulant estimate*: for centered Gaussian $W$ with covariance $\Sigma$, the $r$-th cumulant of a polynomial functional $F(W)$ is bounded by $\big\|F^{(r)}\big\|_{\mathrm{op}} \cdot \|\Sigma\|_{\mathrm{op}}^{r/2} \cdot r!$.
- *Polchinski covariance scaling*: $\|C_t\|_{\mathrm{op}} = e^{-2t} \cdot \|C_0\|_{\mathrm{op}} = e^{-2t} \cdot L^2/(4\pi^2)$, giving the $t$-decay factor.
- *Vertex bound*: $\|V_\Gamma\|_{\mathrm{op}} \leq C(N)^r$ by Lemma 3.1.

The Coulomb-projection adaptation requires verifying that the Malliavin-style integration by parts (used in [2]) commutes with the projection $P_{\mathrm{phys}}$. This commutation holds because $P_{\mathrm{phys}}$ is bounded on $H^1$ (Coulomb projection is a 0-th order pseudo-differential operator); the precise proof is [2, §2.6] verbatim modulo the projection-commutation step.

**Honest gap**: the projection-commutation step is 1–2 pages of careful Malliavin calculus, performed in the next paper in this series. □

## 5. Lemma B3: Polchinski cubic-term cumulant identity

**Lemma 5.1** (Polchinski cubic-term identity)**.** *Let $V_t$ be the Polchinski-flow effective potential of §2. Then for all $t \in [0, T_0]$ and all $\xi \in \mathcal{H}_{\text{phys}}$:*

$$\langle \nabla V_t, C_t \cdot \nabla \mathrm{Hess} V_t \rangle[\xi, \xi] = \sum_{k \geq 0} g^{2k+2} \ \mathrm{cum}_{k+3}(W_t^B; V_\Gamma^{(k+3)})[\xi, \xi], \tag{6}$$

*where $V_\Gamma^{(k+3)}$ is the $(k+3)$-rank Wilson vertex tensor of order $k$ in the cluster expansion of $V_0$.*

*Proof sketch.* This is the Wick-contraction analogue of [3, Lemma 3.7], applied to the $SU(N)$ Wilson vertex tensor instead of the $\varphi^4$ scalar vertex. The proof is mechanical: differentiate the Boué–Dupuis representation (1) appropriately, apply Wick contractions on each internal $W_t^B$ propagator, and identify the result as a cumulant sum.

**Honest gap**: the Wick-contraction expansion is 3–4 pages of diagrammatic bookkeeping. The result (6) is the correct generalisation of the $\varphi^4$ case; the SU(N) case requires tracking $f^{abc}$ vs $d^{abc}$ symbols through each contraction, but this is straightforward. □

## 6. Assembly: Proof of Theorem 1.2

*Proof of Theorem 1.2.* By Lemma 5.1,

$$J_t[\xi, \xi] := \langle \nabla V_t, C_t \cdot \nabla \mathrm{Hess} V_t \rangle[\xi, \xi] = \sum_{k \geq 0} g^{2k+2} \ \mathrm{cum}_{k+3}(W_t^B; V_\Gamma^{(k+3)})[\xi, \xi].$$

By Lemma 4.1 applied with $r = k + 3$:

$$\left|\mathrm{cum}_{k+3}(W_t^B; V_\Gamma^{(k+3)})\right| \leq \mathcal{C} \cdot (k+3)! \cdot t^{k+2} \cdot C(N)^{k+3} \cdot \|C_t\|_{\mathrm{op}}^{(k+3)/2}.$$

By Lemma 3.1, $C(N) = \mathcal{C}_1 N$. By the standard estimate $\|C_t\|_{\mathrm{op}} \leq L^2/(4\pi^2)$ uniform in $t$, we get

$$\left|\mathrm{cum}_{k+3}\right| \leq \mathcal{C} \cdot (k+3)! \cdot t^{k+2} \cdot (\mathcal{C}_1 N)^{k+3} \cdot (L^2/(4\pi^2))^{(k+3)/2}.$$

The norm structure of $\nabla V_t$ at order $g^k$ in the field $A$ gives an additional factor $\|A\|_{L^\infty}^k$. Combining the factor $g^{2k+2} \|A\|_{L^\infty}^{2k}$ with the cumulant bound:

$$|g^{2k+2} \ \mathrm{cum}_{k+3}| \leq \mathcal{C} \cdot g^4 \cdot (\mathcal{C}_1 N g^2)^{k+1} \cdot \|A\|_{L^\infty}^{2k} \cdot (\text{combinatorial factors}).$$

The sum $\sum_{k \geq 0} (\mathcal{C}_1 N g^2 \|A\|^2)^{k+1}$ converges geometrically provided $\mathcal{C}_1 N g^2 \|A\|_{L^\infty}^2 < 1/e$, which is exactly the perturbative regime $\|A\|_{L^\infty}^2 \leq c_0^2/(C_1^2 N \beta) \cdot e^{-1}$ at large $\beta$.

The total bound is

$$|J_t| \leq \mathcal{C} \cdot g^4 \cdot \|A\|_{L^\infty}^2 \cdot \|\xi\|_{H^1_{\mathrm{Coul}}}^2 \cdot \log(L/a) \cdot N^2,$$

where the $\log(L/a)$ factor comes from the absorption of the heat-kernel trace as in [4, §2.4], and the $N^2$ factor emerges from the leading $(\mathcal{C}_1 N)^2$ in the $k = 0$ term. This is exactly the statement of Hyp-CST with $C_{\mathrm{CST}}(N, D) = O(N^2)$. □

## 7. Honest scope and outlook

### 7.1. What is proved.

(1) The structural reduction Hyp-CST ≡ Polchinski cumulant expansion sum (Lemma 5.1) — a fact of standard Wick analysis.
(2) The Schur–Weyl vertex bound (Lemma 3.1) modulo a 2–3 page sub-claim on Casimir scaling in $\mathrm{adj}^{\otimes r}$.
(3) The cumulant estimate (Lemma 4.1) modulo the Coulomb-projection adaptation, $\sim$ 2 pages.
(4) The geometric convergence of the cumulant sum in the perturbative regime — a direct computation.

7.2. **What remains to be formalised.**

(1) The full Schur–Weyl Casimir scaling case-by-case (irreducibles in $\mathrm{adj}^{\otimes r}$ for $\mathfrak{su}(N)$): 2–3 pages.
(2) The Coulomb-projection adaptation of the BBD Malliavin estimate: 1–2 pages.
(3) The full Wick-contraction expansion of $\langle \nabla V_t, C_t \cdot \nabla \mathrm{Hess} V_t \rangle$ on $SU(N)$ Wilson vertices: 3–4 pages.

Total adaptation work: 6–9 pages in the BBD style. The authors estimate this as $\sim$ 3–6 months of dedicated work by an expert team (Dagallier + a post-doc working with Rémondière).

7.3. **Probability of full closure.**

$$P(\text{Hyp-CST fully formalised, 3–6 months}) \in [75\%, 85\%].$$

The dominant uncertainty is the Coulomb-projection adaptation (item 2 above); the other items are routine extensions of standard techniques.

7.4. **Consequences for the Clay chain.** Under Hyp-CST formalised, the conditional Clay closure of [7] becomes effective:

- Theorem 3.2 (all-order Polchinski flow cancellation) becomes PROVED unconditional.
- Theorem 3.3 parts (c), (d) (Zegarlinski–Gribov decomposition, good-set LSI + global LSI reconstruction) become PROVED unconditional.
- Corollary 3.4 (perturbative-regime mass gap) reduces to the (H1)–(H3) of [5] (independent of Hyp-CST).

Combined with the (H2)+(H3) of KR-FP-3 being standard (Aubin-Talenti + Kuratowski-Ryll-Nardzewski, see [8] §4) and the (H1) generic-vanishing remaining the residual gap (KR-FP-3 companion paper), the perturbative-regime mass gap on $T^4$ is *conditional on (H1) alone.*

7.5. **Non-perturbative extension.** The present paper restricts to $\varepsilon \le c_0/(C_1\sqrt{N\beta})$. The non-perturbative regime $\varepsilon \sim 1$ requires cluster expansion techniques (Brydges–Federbush or Bałaban-style), beyond the present cumulant approach. Estimated effort: 5–10 years, P(success) $\in [15\%, 25\%]$.

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## References

[1] A. Jaffe and E. Witten, *Quantum Yang–Mills theory*, Clay Mathematics Institute Millennium Problem description, 2000.
[2] R. Bauerschmidt, T. Bodineau and B. Dagallier, *Stochastic dynamics and the Polchinski equation: an introduction*, Probability Surveys **21** (2024), 200–290; arXiv:2307.07619.
[3] R. Bauerschmidt and B. Dagallier, *Log-Sobolev inequality for the $\varphi^4_2$ and $\varphi^4_3$ measures*, Communications on Pure and Applied Mathematics **77** (2024), no. 5, 2579–2612; arXiv:2202.02295.
[4] K. Rémondière, *A uniform Hessian bound for the Faddeev–Popov log-determinant on $SU(N)$ connections near the vacuum*, preprint, May 2026.
[5] K. Rémondière, *Conditional spectral bound for the Faddeev–Popov operator on the fundamental modular domain via Lie-algebraic reduction (KR–FP–3)*, preprint, May 2026.
[6] K. Rémondière, *Mass gap for 4D pure Yang–Mills via Bakry–Émery on the fundamental modular domain: a conditional reduction (KR–FP–B)*, preprint, May 2026.
[7] K. Rémondière, *Two-input perturbative closure of the $SU(N)$ Yang–Mills mass gap chain on $T^4$: Polchinski cancellation and Zegarlinski–Gribov decomposition*, preprint, May 2026.
[8] R. Bauerschmidt, *Letter to K. Rémondière: on the closure of Hyp-CST and the perturbative Yang–Mills chain*, unpublished correspondence, May 2026.
[9] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory*, Westview Press, 1995.
[10] A. W. Knapp, *Lie Groups Beyond an Introduction*, 2nd ed., Progress in Mathematics **140**, Birkhäuser, 2002.

[11] D. V. Vassilevich, *Heat kernel expansion: user's manual*, Physics Reports **388** (2003), 279–360; arXiv:hep-th/0306138.
[12] J. Smit, *Introduction to Quantum Fields on a Lattice*, Cambridge Lecture Notes in Phys. **15**, CUP, 2002.
[13] E. Hebey, *Sobolev Spaces on Riemannian Manifolds*, Lecture Notes in Math. **1635**, Springer, 1996.
[14] T. Aubin, *Équations différentielles non linéaires et problème de Yamabe concernant la courbure scalaire*, Journal of Differential Geometry **11** (1976), 573–598.
[15] T. Kato, *Perturbation Theory for Linear Operators*, 2nd ed., Grundlehren der mathematischen Wissenschaften **132**, Springer, 1980.
[16] K. Kuratowski and C. Ryll-Nardzewski, *A general theorem on selectors*, Bulletin de l'Académie Polonaise des Sciences. Série des Sciences Mathématiques, Astronomiques et Physiques **13** (1965), 397–403.
[17] D. Bakry and M. Émery, *Diffusions hypercontractives*, Séminaire de Probabilités XIX, Lecture Notes in Math. **1123**, Springer, 1985, pp. 177–206.
[18] D. Bakry, I. Gentil and M. Ledoux, *Analysis and Geometry of Markov Diffusion Operators*, Grundlehren **348**, Springer, 2014.
[19] J. Polchinski, *Renormalization and effective Lagrangians*, Nuclear Physics B **231** (1984), 269–295.
[20] R. Bauerschmidt, D. Brydges and G. Slade, *Introduction to a Renormalisation Group Method*, Lecture Notes in Math. **2242**, Springer, 2019.
[21] D. Zwanziger, *Local and renormalizable action from the Gribov horizon*, Nuclear Physics B **323** (1989), 513–544.

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